



Advances in Artificial Intelligence-Based Medical Devices for Healthcare Applications

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Abstract

Medical devices play a crucial role in modern healthcare, providing innovative solutions for diagnosing, preventing, monitoring, and treating ailments. Artificial Intelligence is transforming the field of medical devices, offering unprecedented opportunities through diagnostic accuracy, personalized treatment plans, and enhancing patient outcomes. This review outlines the applications of artificial intelligence-based medical devices in healthcare specialties, especially in dentistry, medical imaging, ophthalmology, mental health, autism spectrum disorder diagnosis, oncology, and general medicine. Specifically, the review highlights advancements such as improved diagnostic accuracy, tailored treatment planning, and enhanced clinical outcomes in the above-mentioned applications. Regulatory approval remains a key issue, where medical devices must be approved or cleared by the United States Food and Drug Administration to establish their safety and efficacy. The regulatory guidance pathway for artificial intelligence-based medical devices is presented and moreover the critical technical, ethical, and implementation challenges that must be addressed for large-scale adoption are discussed. The review concludes that the intersection of artificial intelligence with the medical device domain and internet-enabled or enhanced technology, such as biotechnology, nanotechnology, and personalized therapeutics, enables an enormous opportunity to accelerate customized and patient-centered care. By evaluating these advancements and challenges, the study aims to present insights into the future trajectory of smart medical technologies and their role in advancing personalized, patient-centered care.

Keywords Artificial intelligence (AI) · Deep learning · FDA · Machine learning · Medical device

Artificial Intelligence in Medical Devices

Medical devices are essential in modern healthcare, assisting in patients' diagnosis, treatment, monitoring, and rehabilitation [1]. People around the world benefit from medical devices by improving their health and enhancing their quality of life. The medical device industry has consistently changed from basic tools like bandages and syringes to sophisticated devices like pacemakers and magnetic

resonance imaging (MRI) machines to satisfy the expanding needs of the healthcare sector [2, 3]. Recently, artificial intelligence (AI), a type of software that examines data from various devices facilitates early detection and accurate diagnosis of health issues, and provide effective treatments, ultimately leading to improved patient health [4, 5]. The importance of medical devices and its related software cannot be overemphasized. All medical devices and their software are regulated by the United States Food and Drug Administration (FDA). The medical device development process involves multiple stages from concept to market availability, but the development and regulation of medical devices are not without their challenges. For effective and safe products assurance, manufacturers have to handle a complex web of regulations, risk management procedures, and safety standards [2, 6]. Another challenge in the medical device industry is the rapid pace of technological advancement. New materials and techniques, and digital technologies are constantly emerging, requiring manufacturers to adjust their development processes to stay ahead of the curve. For patient safety

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and data privacy protection, concerns with device compatibility, interoperability, and data security may also arise from this quick evolution [7]. Moreover, strong clinical evidence is required to reinforce the effectiveness and safety of medical technologies. AI and many other software have been adopted in the healthcare and are very promising thus far.

The FDA has implemented more stringent requirements for clinical data for medical devices, which can increase the time and cost of device development. Despite these challenges, the medical device industry has witnessed remarkable advancements in recent years. The designing, production, and usage of medical devices have been completely transformed by the incorporation of AI, machine learning (ML), and the internet of things (IoT). These innovations have led to the development of more personalized, adaptive, and connected devices that can provide real-time data and improve patient health. AI has become an increasingly important tool in the development and evaluation of medical devices [8, 9]. AI algorithms can automate and streamline various aspects of the medical device lifecycle, from literature review to clinical testing and regulatory approval. During medical device development, AI can assist with tasks like optimizing device design, predicting device performance, and automating testing procedures. AI-powered simulations can model device behaviour and interactions with the human body, reducing the need for costly physical prototyping and animal studies. John McCarthy coined the term AI in 1955 as making intelligent machines [10]. AI now entails employing computers to analyze intricate medical data in order to simulate intelligent behaviour with little assistance from humans [10, 11]. It is estimated that the use of AI in the medical field will increase from \$1.3 billion to \$10 billion by 2024 globally, signifying a 40% yearly growth rate [12]. Computers may eventually match human intelligence if advancements in software development,

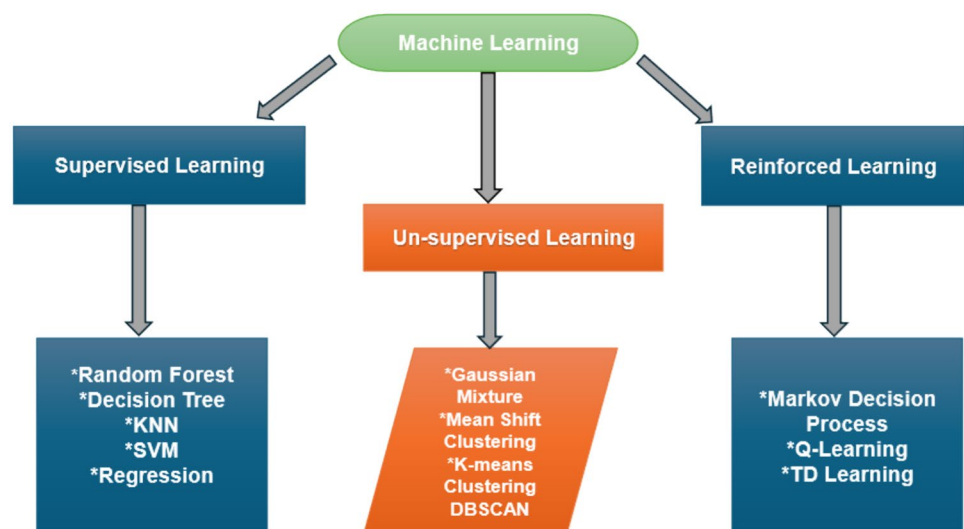
technological speed, and capacity are sustained. AI applies novel concepts and solutions to solve complex challenges. Radiology [13], oncology [14], ophthalmology [15], and many more medical fields have been demonstrated to benefit from AI/ML-based technology. The number of approved AI/ML-based medical devices by the FDA increased from 67 in 2018, to 77 in 2019 [16]. As of May 2024, the FDA has approved 882 AI-enabled devices. AI tools utilized in medical device includes ML [17], deep learning (DL) [18, 19] and natural language processing (NLP) methods [20, 21]. The ML technique is a subset of AI whereas, DL technique is a sub-domain of ML but an advanced form of neural network. The algorithms of all these subsets of AI are widely employed in speech recognition, computer vision, drug discovery, and genomics [22, 23]. Each of the subsets of AI are discussed further.

Machine Learning

ML assimilates knowledge from data over time and use this information to make predictions upon the introduction of new datasets [24, 25]. ML tools can turn large volumes of data into useful information by providing reports based on historical data and establishing models to predict potential outcomes [26]. The three (3) main categories of ML with their characteristics include supervised learning, unsupervised learning, and reinforcement learning [27]. Figure 1 below shows the main ML types and their features.

K-nearest neighbour (KNN) is one of the simplest and most essential supervised learning classifier which uses proximity to make classifications [28]. The number of nearest neighbours to a new unknown variable to be classified is represented as support vector machine (SVM) is utilized in image, hypertext, text segregation and classification problems. Utilized in domains such as face recognitions etc. [29].

Fig. 1 Classification of machine learning algorithms



K-means clustering where k is assumed as certain number of clusters is an unsupervised learning algorithm used in resolving clustering problems. Density based spatial clustering of applications with noise (DBSCAN) is utilized for clustering spatial databases with noise. It can uncover clusters of arbitrary shape and also handles outliers efficiently [30]. Quality learning (Q-learning) is form of model-free reinforcement learning widely utilized in statistics [31]. Temporal differencing (TD)-learning is used for value function learning [32].

In supervised ML, the machine operates using selected training data sets. The input and output data are both distinctly listed. Training data sets are taken as reference or accepted information for learning, and a regression or classification model is constructed that can respond to unknown information [33]. The disadvantage of supervised learning is that when the volume of the data is large it becomes very costly. Classifying spam emails from your inbox, fraud detection, image classification, and other real-world uses are some examples. Regression analysis and other decision tree-based analytic techniques like the random forest approach are examples of utilizing supervised learning [34–36]. Unsupervised learning is the type that does not require training data to establish the true answer [37]. Unsupervised learning is when the model discovers the hidden patterns or groups the data on its own without any human intervention. The disadvantage of unsupervised ML is that since the input is not labelled, it may result in an inaccurate output. Some commonly used unsupervised learning approach includes, Restricted Boltzmann Machines (RBMs), and Generative Adversarial Networks (GANs). GANs are largely used for data enhancement purpose [38]. A typical unsupervised learning algorithm is clustering [39, 40]. Reinforcement learning is a learning method where a software agent interacts with an unknown environment, selects actions, and progressively discovers the environment dynamics [41]. Reinforcement learning uses trial and error to create algorithms that learn the most rewarding behaviours and choices and makes decisions [42, 43]. An agent is placed in an environment in reinforcement learning. The agent needs to act and obtain the greatest number of rewards. In this case, the computer must make choices based on experience and rewards because the path is not predetermined. Numerous significant spheres of real life have seen the successful application of reinforcement learning. However, there is a unique kind of learning known as semi-supervised learning. Semi-supervised learning lies in between supervised and unsupervised machine learning. In supervised learning, labelled data are used to train the machine; in unsupervised learning, unlabelled data are provided, and the machine's task is to classify the data according to similarities and differences. The introduction of semi-supervised learning addresses the drawbacks of both supervised and unsupervised learning.

Deep Learning

Deep learning (DL), a subfield of ML, is revolutionizing the way we approach complex tasks. It involves training neural networks to perform specific tasks by learning intricate concepts from simpler ones [44]. With the use of sophisticated neural network algorithms inspired by the neural architectures of the human brain [22], deep learning approaches can process large amounts of unprocessed, complex data with astounding speed. The artificial neural network (ANN), a computer software that mimics how the human brain functions, is the central component of deep learning [45]. The input, hidden, and output layers make up the three foremost layer categories of the neural networks. The input layer receives the raw data, and processes it through several “hidden” layers, adjusting the weighting of the information to boost the output's accuracy rate [46]. Unlike traditional ML methods, DL algorithms can learn directly from raw data without human guidance, uncovering latent relationships and patterns that may have been overlooked [47]. DL has had an extremely positive impact on the medical industry. In 2016, a deep learning-based algorithm was first introduced into medicine where DL demonstrated accuracy comparable to ophthalmologists in grading diabetic retinopathy from fundus images [48], marking a significant milestone in the field. The following year, reports emerged of a dermatologist-level algorithm for classifying skin cancer into benign or malignant [49], as well as a pathologist-level algorithm for detecting metastatic lesions of breast cancer from tissue sections of lymph nodes [50]. These achievements highlight the potential of DL in revolutionizing medical diagnostics and treatment. DL has been described as less robotic than traditional ML. To be considered “deep,” an ANN must have more than one hidden layer [51]. These layers are made up of nodes that combine data input with a set of coefficients (weights) that amplify or dampen that input in terms of its effect on the output. Two of the most common forms of DL models for supervised learning are convolutional neural networks (CNNs) and recurrent neural networks (RNNs). CNNs are particularly adept at processing image and signal data, thanks to their ability to create local connectivity across an image or signal. These local connections, combined with non-linear activation functions, transform the representation into a higher, slightly more abstract representation. Conversely, RNNs are made to handle sequential input, like speech or text, by integrating feedback loops that let information linger and affect computations thereafter. While DL has already made significant strides in various fields, an important caution in DL is that the hidden layers within ANNs can render the output harder to interpret [52]. That said, deep neural network topologies and continuous research continue to push the envelope of what is feasible. As we delve deeper into the realm of DL, we can expect

to witness even more groundbreaking applications that will transform industries and improve our understanding of the world around us [53].

Natural Language Processing

Natural language processing (NLP), a subfield of AI and linguistics, is devoted to make computers understand the statements or words written in human languages [54]. For all sophisticated, and challenging tasks related to the language, such as machine translation, question answering, summarization, NLP is the preferred process. The NLP method produces information from unstructured data such as clinical data to develop into structured medical data. The purpose of NLP is to transform texts into structured data that are readable by machines so that ML can examine it [55, 56]. Two components natural language understanding (NLU) and natural language generation (NLG) are used in natural language processing [57].

Recent advancements, as illustrated in Table 1, emphasize the transformative prospects of integrating artificial intelligence into medical devices. Artificial Intelligence has the potential to transform healthcare by providing high-accuracy diagnostics, personalized treatment regimens, real-time monitoring, and prediction analytics. These capabilities significantly influence healthcare decision-making processes and patient outcomes across various medical specialties. Despite these advantages, the development and deployment of artificial intelligence-based medical devices are fraught with challenges. The rest of the review is structured in the following way. This review explores the current state of artificial intelligence integration in medical devices, focusing on applications such as dentistry, medical imaging, glaucoma, autism spectrum disorder, mental health, oncology, and general medicine. Furthermore, the regulatory pathways workflow guidelines are illustrated with a focus on certain AI-based medical devices approved by the FDA in various medical specialties. Finally, an emphasis on the key challenges of the AI-based medical devices is reported.

Table 1 Recent advances in AI-based medical devices

Author(s)	Summary
Srivastava et al. [58]	AI algorithms were used to predict the behaviour of drug-target proteins and their physicochemical properties, restructuring the drug discovery process. The capability reduced the time and cost associated with bringing new drugs to market by enabling virtual screening and modeling of potential drug candidates. The integration of AI with technologies like 3D printing facilitated the creation of personalized medications tailored to individual patient needs [58]
Chen et al. [59]	The integration of AI algorithms with medical sensors significantly improved the processing of large datasets, facilitating early disease prediction and timely diagnosis as per this work. The work discussed the development of AI-enabled medical sensors for various applications, including off-body detection, near-body monitoring, disease prediction, and clinical decision support systems [59]
Fraser et al. [60]	The authors emphasized that the level of clinical evidence required for AI-based medical devices should be determined based on the specific application and associated risks. Factors such as accountability, transparency, and interpretability are crucial in this assessment. The paper advocated for the development of common standards for the clinical evaluation of high-risk AI applications. Such standards would benefit regulators, manufacturers, clinicians, and patients by ensuring transparency in evidence and performance metrics [60]
Doctor et al. [61]	This work reported that, incorporating AI-based troubleshooters in medical device manufacturing holds significant promise for improving operational efficiency, product quality, and compliance with industry regulations [61]
Dennis et al. [62]	Dennis and co examined the performance enhancement of an AI-driven diagnostic tool designed to assist in identifying autism spectrum disorder (ASD) in children. This optimization was achieved through a predetermined change control plan (PCCP), aligning with the FDA's guidelines for managing iterative updates in AI-enabled medical devices [62]
Eui Jin Hwang et al. [63]	They addressed the integration of AI-based software as medical devices (AI-SaMDs) into clinical chest radiography. This work underscored that AI-SaMDs should serve as supportive tools for radiologists, enhancing diagnostic accuracy without replacing human judgment. Radiologists were vehemently encouraged to critically assess AI outputs within the broader clinical context [63]
Badnjević et al. [64]	This work utilized AI to enhance diagnostic accuracy in analyzing cellular characteristics for an In vitro diagnostic platform. Clinical engineering tools that applied ML was used to predict equipment failures, thereby improving maintenance strategies and patient safety [64]
Arima and Kano [65]	The authors identified a “double-layered” regulatory issue, where differences existed both in medical device regulations and in AI-specific guidelines across countries. This disparity was said to complicate the global development and deployment of AI-based medical devices [65]
Robin et al. [66]	The paper determined that the synergy of AI, wearable devices, and IoT applications has significantly contributed to the prevention, screening, and treatment of COVID-19. These technologies not only improved patient outcomes but also paved the way for more resilient and responsive healthcare systems in the face of global health crises [66]

Artificial Intelligence (AI) in Health Care

Artificial intelligence in healthcare is a vast domain that utilizes data-driven techniques and advanced algorithms to enhance patient diagnosis, optimize therapy regimens, and streamline medical procedures. AI in medical devices is a practical application of AI in health care, where advanced algorithms are integrated directly into the digital tools and hardware's utilized in clinical environments. Collectively, they are transforming the way healthcare is delivered by making it more personalized, proactive, and efficient. Despite the generativity of human language, a computer algorithm's capacity to automatically deduce the meanings of underlying words is a significant technological achievement and is crucial for applications in mental healthcare due to considerable raw input data in the form of text (e.g., clinical notes; other written language) and conversation (e.g., counselling sessions) [52]. NLP can improve hospital triage systems, predict patient outcomes, and create diagnostic models that identify chronic diseases in their early stages when used with medical notes [67]. In critical care, where patient death prediction is a regular practice and there is an abundance of patient data to evaluate, these applications might be especially helpful. NLG is a function of NLP, which makes it possible for patients to interact with chatbots to ask inquiries and obtain pertinent information in addition to NLU. NLP is used daily by every smartphone user, and has contributed to the development of language translation, personal digital assistants, and voice-controlled home automation systems [68, 69].

With the current healthcare facing a lot of challenges in collecting, analyzing and applying structured and unstructured data to diagnose and treat diseases, AI systems with their data mining and recognition abilities will provide several benefits over clinical and conventional analytic decision-making techniques [70]. With the help of AI, algorithms obtained will be more accurate and clearer that will help clinicians to provide unprecedented diagnosis treatment and care to the patients [71]. General medical diagnosis involves three major steps. Firstly, observation and examination of the patient, collection of patient information, interpretation of the information obtained based on clinician's knowledge and experience. Secondly, formulation of a diagnosis and finally, therapeutic plan by the physician. AI technology facilitates the rapid extraction of data from a large patient pool and supports real-time inference for health risk assessment and prediction of health outcomes [72]. Big data from smartphone apps, wearable technology, social media, environmental and lifestyle factors, sociodemographic, omic data (genomics, metabolomics, proteomics, radiomics, etc.), and data from standardized electronic health records or precision

medicine platforms can all be analyzed using AI technology [73, 74]. According to Naylor, there are seven factors influencing the use of AI and DL in healthcare: (1) the benefits of digital imaging over human interpretation; (2) the digitization of health records and data sharing; (3) the flexibility of deep learning for the analysis of heterogeneous data sets; (4) the potential of deep learning for the generation of research hypotheses; (5) the potential of DL to expedite clinical workflows and give patients more power; (6) the quick spread of both open-source and proprietary DL applications; and (7) the sufficiency of current basic DL technology to yield better results as data sets grow. The other variables also apply to other AI techniques; only elements 3, 4, and 6 are unique to DL [75]. Despite AI's potential, it remains challenging. New regulations are needed to keep up with the quickly evolving field of technology as consumer technologies like blockchain, AI, 3D bioprinting, cloud computing, and cloud computing find their way into the healthcare industry. A more thorough regulatory pathway for innovations in this field and new instructions offering advice on the technical aspects of producing 3D-printed medical devices were recently issued by the FDA in response to the growing trend of 3D-bioprinted medical technologies [76].

AI Analytical Tools in Healthcare

The development of AI tools intended to help the clinician in diagnosis formulation, therapeutic decisions and the prediction of outcome has become relevant. These tools help healthcare professionals carry out their daily responsibilities by supporting tasks that require the processing of knowledge and data [77]. The AI tools commonly used in healthcare include ANNs, fuzzy expert systems, and genetic algorithms (GA) [78]. ANN is the most popular AI tool in medicine [79]. It is made up of “neurons,” which are networks of highly interconnected computer processors that can process input and express knowledge in parallel. They are a very appealing analytical tool in medicine because of their capacity to manage inaccurate information, interpret non-linear data, and learn from the past. It has been demonstrated that ANN was utilized in predicting survival in patients with breast and colorectal cancer [80]. Fuzzy logic is a cognitive science that recognizes and uses real-world incidents. Since most medical data are inherently imprecise and medicine is fundamentally a continuous domain, fuzzy logic is a data handling methodology that allows for ambiguity, making it particularly well-suited to medical treatments. It effectively and computationally conveys the notion of fuzziness. Fuzzy logic has been investigated for use in the detection of acute leukemia [81], breast [82] and pancreatic [83] cancer. GA is a computational performance based on biological evolution using the mechanism of the survival of the fittest in solving

real-world problems [84]. They work by creating many random solutions to the problem at hand. Subsequently, the population of multiple solutions will develop across successive generations until a workable solution to the problem is found. The population grows by adding the best solutions and weeding out the worst ones [85]. By continuing this process, the population will continue to improve, and new solutions will be produced. It uses methods including mutation, inheritance, selection, and crossover to try to solve the problem better. The fact that GA is centered on problem solving as opposed to the idea of analytical relations is by far its greatest advantage over conventional techniques [86]. It is based on simple criteria and is therefore easy to use even if it is a powerful optimization tool. Critically sick patients have had their outcomes predicted using GA principles [87].

AI Applications in Healthcare Specialties

AI in Dentistry

The logical procedures that dentists employ is the same applied by many AI/ML models. AI techniques possess great potential in identifying and classifying oral cavity lesions, a task that is typically beyond observation by human sight [88]. Fuzzy logic, convolution neural networks (CNNs), ANNs, and GAs are some of the AI technologies used in dentistry [89, 90]. To identify proximal caries, Choi et al. suggested combining a number of image processing methods with CNN [91].

The main benefit of these technologies is their ability to resolve issues that are too complicated in traditional approaches. In research by Kim et al., a model was developed using ANN to predict toothache in relation to the daily toothbrushing frequency, time spent brushing, use of dental floss, toothbrush replacement schedule, scaling, and other variables including nutrition and activity [92]. GA-based strategy was utilized for detecting dental caries early and preventing decay severity [93]. The goal of fuzzy logic was to mimic human reasoning abilities that function with ambiguous concepts. Hererra and associates employed a fuzzy logic-based approach to anticipate the change in colour after bleaching based on the original chrome values of the teeth [94]. AI has been utilized in different dentistry specialties. In orthodontics, the most recent revolution was AI driven customized orthodontic treatment used in assessing craniofacial and dental abnormalities [95]. In Forensic Odontology, AI was employed for analyzing bite marks and predicting mandibular morphology [96]. AI in dental radiography has been developed to perform image-based tasks such as classification, detection and segmentation as shown in Fig. 2.

In Oral and Maxillofacial Surgery, the development of robotic surgery is the biggest use of AI in oral surgery.

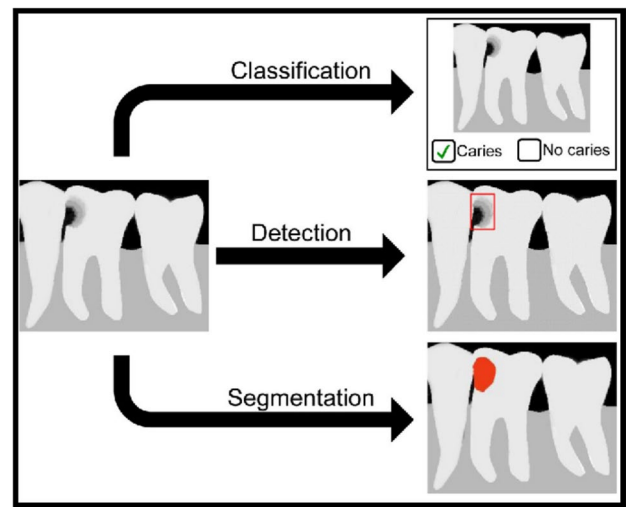


Fig. 2 Most frequent computer vision tasks with an example of dental caries recognition [97]. © 2021, British Institute of Radiology

Several robotic surgeons with growing efficiency, now carry out semi-automated surgical procedures under the supervision of a skilled surgeon [98]. In Prosthetic Dentistry, the use of computer-aided technology for precise fit of prosthesis is yet another breakthrough of AI in dentistry [90]. In periodontics, ANN can also effectively be used in categorizing patients into acute periodontitis and chronic periodontitis based on their immune response profile [99]. AI has drastically changed the dental chair from conventional hydraulic chair to electrical and fully automatic dental chair operated through sensors. A recent advance is a voice command operated dental chair which needs no physical effort from the dentist. The future is not very far when a dental chair can calculate patients' weight, vital signs and anxiety as the patient sits on the dental chair for treatment. Despite the promising results of the presented AI models, it is still necessary to verify their generalizability and reliability using appropriate external data obtained from freshly enlisted patients or accumulated from other dental facilities.

AI in Radiology/Medical Imaging

Medical imaging is the healthcare industry's largest and fastest-growing data source [100]. It accounts for 90 percent of all healthcare data and more than 97 percent of it goes unanalyzed or unused. Radiologists are under increasing pressure to read quickly and accurately the increasing number of scans recently [101, 102]. In medical imaging, attempts have been made to introduce AI into endoscopic, MRI, CT, ultrasound and pathological imaging. Services from CT scans to X-rays diagnose, treat and monitor illnesses and conditions faster and more accurately than ever before. Prior to now, the processing of the enormous

amounts of medical imaging data resulted in longer wait times from image acquisition, diagnosis and treatment [103]. AI development has significantly decreased wait times to obtain medical imaging results. For instance AI has been used to identify and distinguish between four different diseases in chest radiographs, including primary lung cancers and metastasis, active pulmonary tuberculosis, pneumonia, and pneumothorax [104–106]. AI's effectiveness as a second reader in conjunction with a radiologist in enhancing lesion localization has been demonstrated as shown in Fig. 3 [107]. Another study showed that AI-based computer-aided design software flagged fewer false positives per image in digital mammograms than FDA-approved computer-aided design software, with no reduction in sensitivity [108]. AI has been used for the treatment planning of brain tumors and

the identification of catheters and tubes in pediatric x-ray images [109, 110].

AI in Ophthalmology (Glaucoma)

Phthalmology generally involves the diagnosis and treatment of eye disorders. Glaucoma is considered as a group of eye diseases characterized by optic neuropathy, a specific pattern of visual field defect and increased intraocular pressure. Compression of the retina and optic nerve as a result of an increase in intraocular pressure results in permanent loss of eyesight [112]. Glaucoma is known as the major cause of irreversible blindness worldwide [113]. It is estimated that approximately 64.3 million individuals are affected by this disease and this number is expected to reach approximately

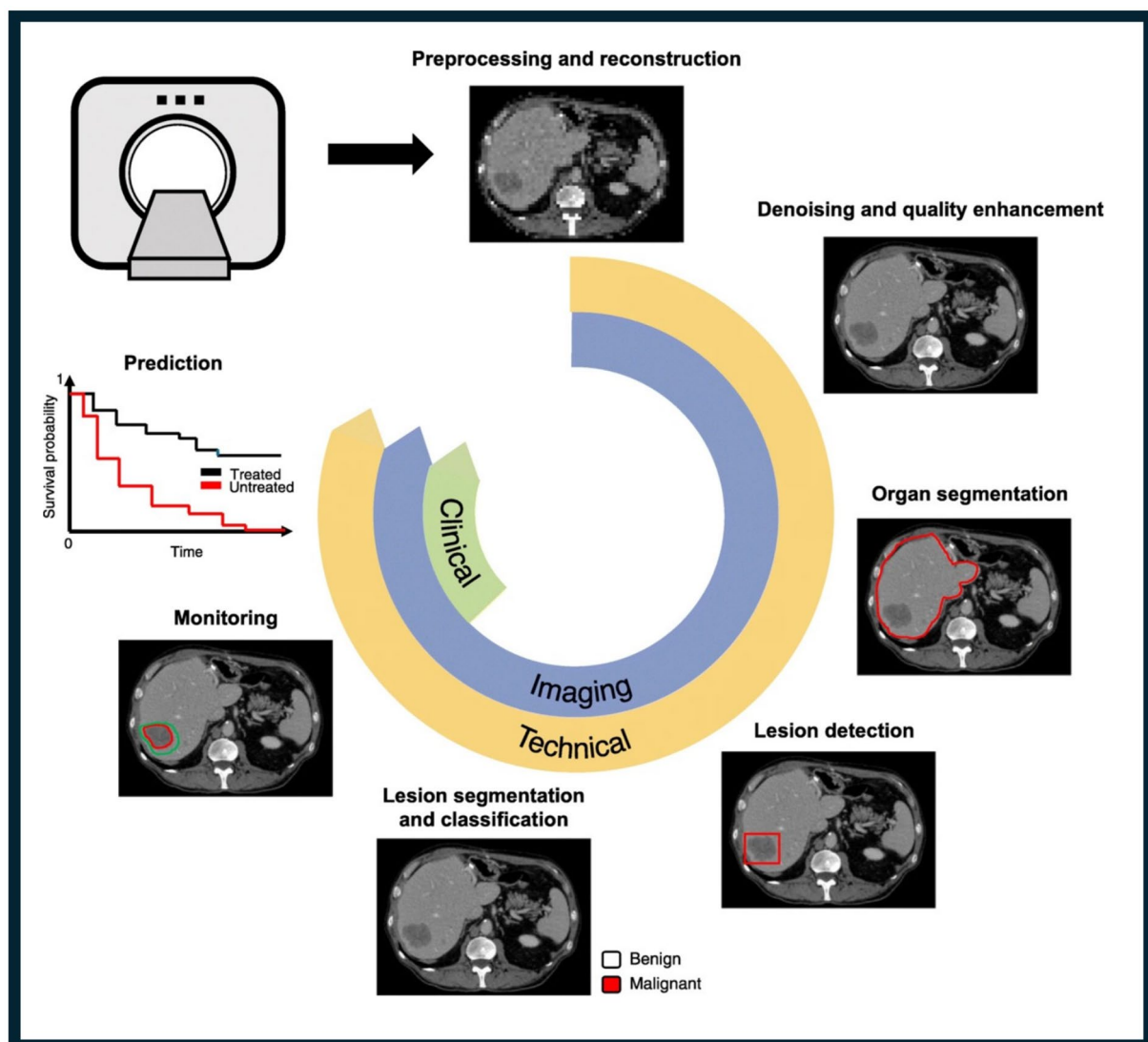


Fig. 3 Potential use of DL methods in medicine. Lesion identification, classification, and long-term follow-up can be used to inform tasks like treatment response monitoring and survival prediction [111]. © 2020, Emmanuel Montagnon et al.

112 million by 2040 [114]. Detection of glaucoma disease in its early stages may prevent blindness. Undiagnosed glaucoma cases still account for more than 50% of the cases worldwide [115]. In ophthalmology, clinical data can be analyzed by using NLP and ML/DL (deep learning) techniques. There are several clinical methods available for glaucoma detection but the increase in population and rising levels of obesity gives rise to the situation that demands more ophthalmologists. This becomes a restrictive factor, necessitating the need to develop techniques which can recognize characteristics of pathological cases automatically. Diagnosis of glaucoma is related to the assessment of the optic disc and the adjacent retinal nerve fibre layer by a trained specialist. A black box problem has been a challenge in using AI to address the diagnosis of glaucoma. This is because, once the software has finished learning, it is often not apparent to the programmer how exactly the ML program that they had written generates the required output. This is called the black box problem and the reason why AI-generated reports lost trust. Newer AI software opens up the black box using an Integrated Gradients Explanation algorithm to show a heatmap or attention map as Google researchers demonstrated [116] and shown in Fig. 4.

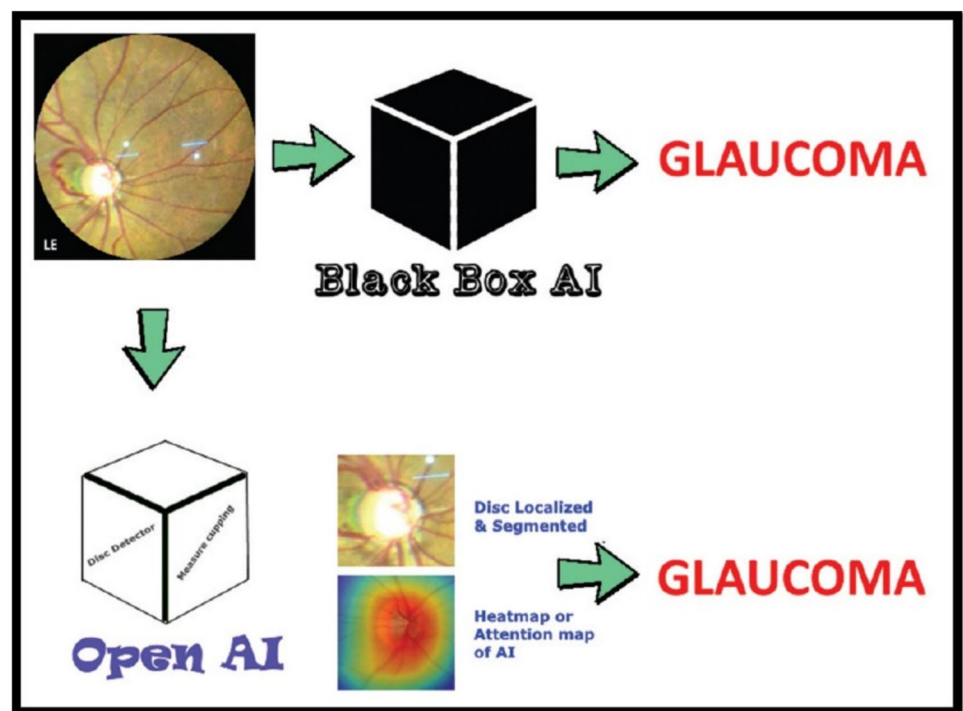
Conventional methods for detection of glaucoma have higher chances of being misdiagnosed. Therefore, there is a need for an automated system which prevents misdiagnosis and makes correct use of the available medical data. As electronic medical records are becoming mainstream along with advancement in digital imaging and data storage, it

is important to appropriately utilize clinical data and digital images more than ever before. Currently, glaucoma is diagnosed using various test methods, like measurement of intraocular pressure, visual field tests, fundus and optical coherence tomography (OCT) imaging. Out of all the test methods OCT visualizes and quantifies structures in the eye. These are the retinal nerve fibre layer (RNFL), cup-to-disc ratio (CDR) which changes with the progression of the disease. AI along with OCT imaging creates an algorithm for detection as well as diagnosis of glaucoma with the recent example shown in the study by Lee et al. using deep learning technique for automated segmentation of macular edema in optical coherence tomography. The eye being the only organ that allows direct visualization of small vessels, AI analysis of the fundus image was used to predict the risk of systemic diseases of the heart and kidney as shown in Fig. 5. Table 2 summarizes the techniques for glaucoma analysis.

AI in Autism Spectrum Disorder Diagnosis

Autism spectrum disorder (ASD) is one of the fastest-growing disorders, accounting for 1.9 percent of the population in the United States [121]. Approximately 27% of children with ASD are still undiagnosed at age 8 [122]. The enormous increase in demand for ASD evaluations, which has outpaced specialist capacity and resulted in prolonged wait times, is one factor contributing to diagnostic delay [123–126]. Access and diagnosis have also been hindered by gender, race, and socioeconomic biases [127, 128]. The

Fig. 4 Representation of the black box problem in AI. Typical AI does not show how the analysis works. By modifying the internal working, “Open” AI shows the segmentation and heatmap by which the image was analyzed and arrived at a diagnosis [117]. © 2019 Kerala Journal of Ophthalmology. Published by Wolters Kluwer-Medknow



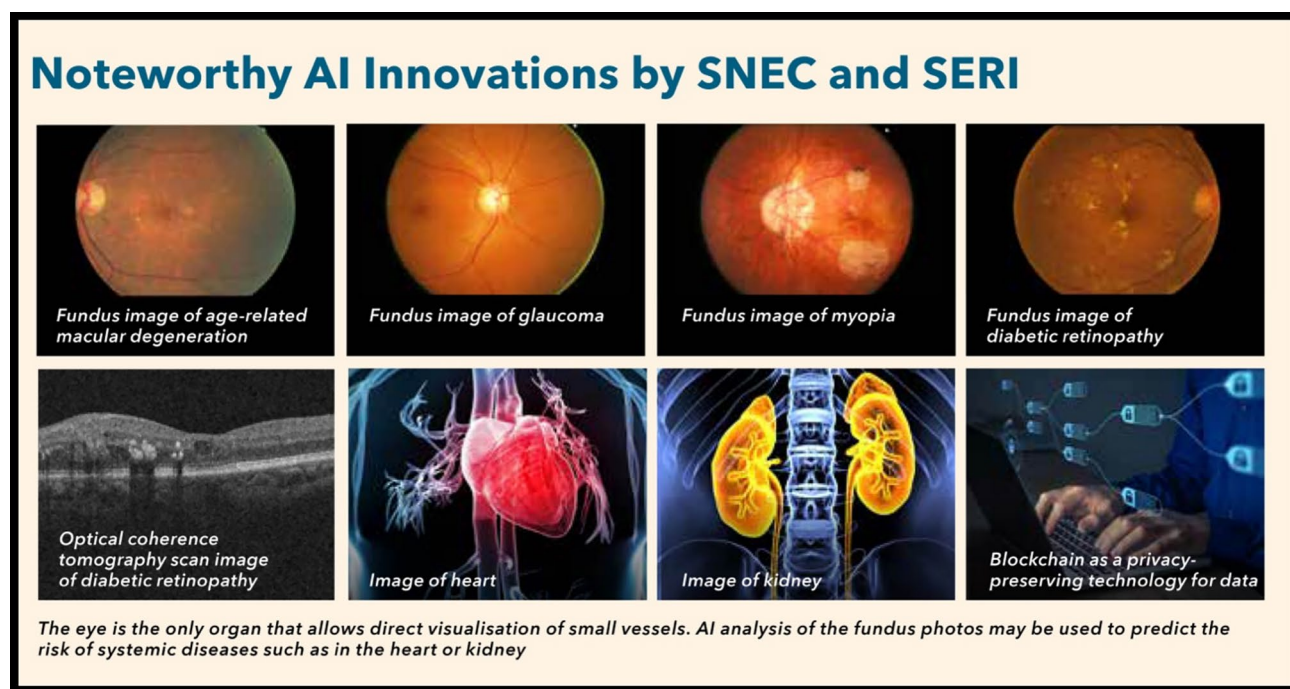


Fig. 5 AI analysis of the fundus image to predict the risk of systemic diseases of the heart, kidney [118] Wong and Bressler used deep learning technology for screening of diabetic retinopathy [119]. Deep

convolutional neural networks was utilized to identify glaucoma, according to Chen et al. [120]

Table 2 Imaging procedure for the diagnosis of glaucoma

Technique	Function
Fundus image	Gives an extended view of the retina
Optical coherence Tomography	Utilizes light waves to take cross-section pictures of the retina
Slit-lamp photography	Gives a stereoscope detailed extended view of the anterior segment
Visual field analysis	This shows the shape and size of the field of view

assessment takes time, and families may have to wait up to 18 months between the preliminary screening by their healthcare provider (HCP) in the primary care setting and the final diagnosis by the specialist [129]. Standard diagnosis tools used in primary care settings may miss many cases of ASD. The difficulty of making an accurate ASD diagnosis may also contribute to primary care HCPs' reluctance to diagnose. Moreover, attention-deficit hyperactivity disorder (ADHD), intellectual disability, speech and language delay, and a number of psychiatric diseases can co-occur with or share many overlapping symptoms with ASD in children [130]. In a time-constrained clinical interaction, providers must also assess children whose conduct may not match that of their home setting or who may demonstrate newly reactive behaviour as a result of environmental changes [131]. No diagnostic tool had FDA market permission prior to June 2, 2021, which would have made it easier to diagnose ASD in a primary care context (FDA). In groups where ASD is less common, like primary care, overdiagnosis is

also a possibility [132]. To improve primary care health care providers' ability to diagnose ASD and send complex cases for expert evaluation, creative diagnostic tools for use in the primary care context are immediately needed. AI-based solutions have the potential to improve a number of elements of ASD care, according to recent research [133] (see Fig. 6).

Validation of AI-based ASD

To verify the accuracy of an AI-based software as a medical device intended to assist primary care healthcare providers (HCPs) in diagnosing ASD, a double-blind, multi-site, prospective, active comparator cohort research was implemented. In order to generate an ASD positive, ASD negative, or indeterminate output, the device uses a gradient boosted decision tree machine learning algorithm to combine behavioural features from three different inputs (a caregiver questionnaire, analysis of two short home videos, and an HCP questionnaire), as shown in Fig. 11. Before the trial started,

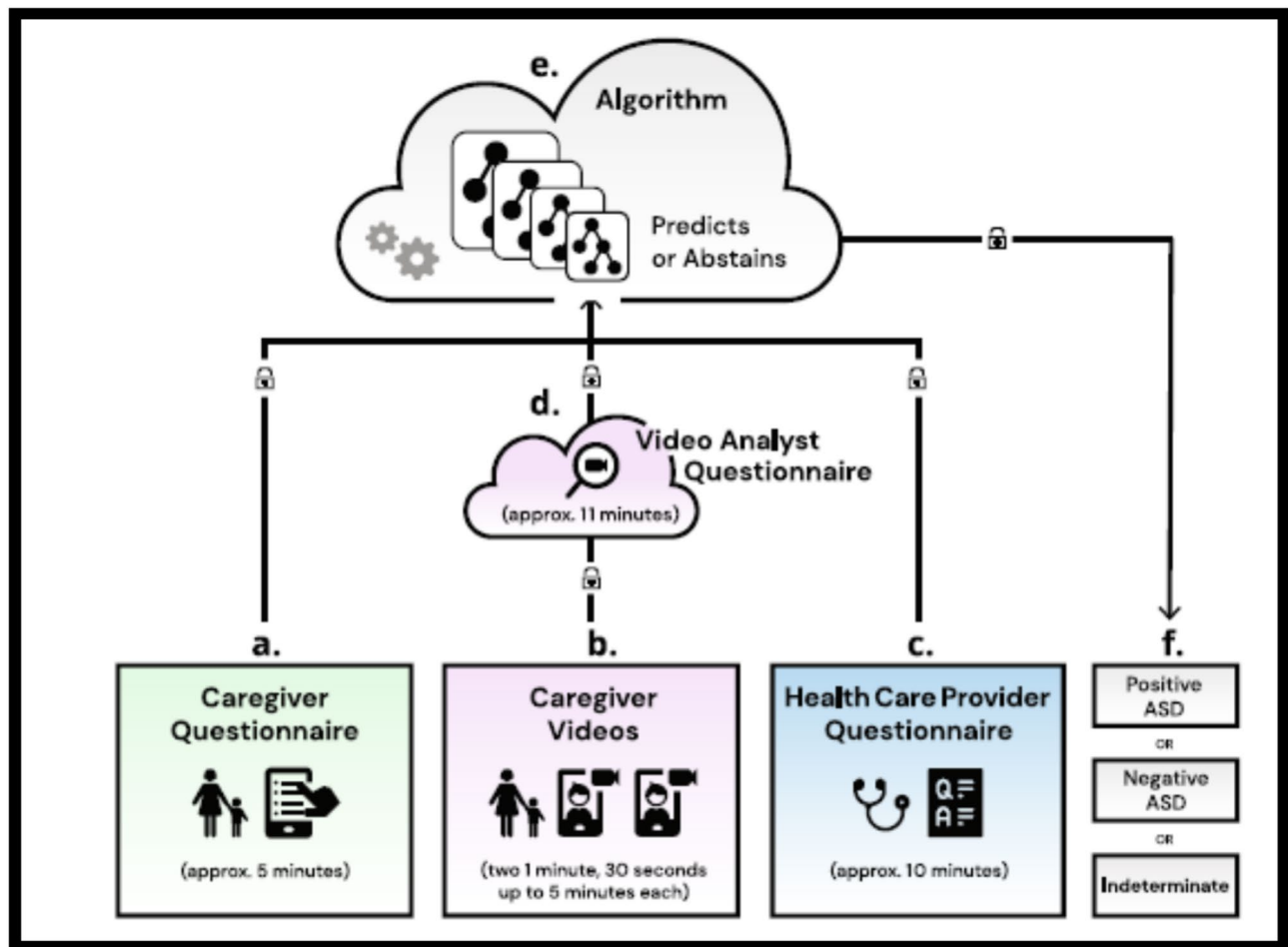


Fig. 6 A visual representation of the device and its main parts. In about five minutes, the caregiver will: **a** use a smartphone to complete a brief questionnaire; **b** upload two brief home videos of their child—one lasting one minute, thirty seconds, or up to five minutes—for evaluation by skilled video analysts; and **c** their primary care physician, or another qualified healthcare provider, will independently respond to a set of brief clinical questions in about ten minutes. The expert analysts receive these inputs safely, and they take about

11 min to extract the video features. The inputs from the video analyst, healthcare provider, and caregiver are merged into a mathematical vector for categorization and machine learning analysis. An “ASD positive,” “ASD negative,” or “no result (indeterminate)” output is produced by the device. The uncertain result shows that at that point, the information was not specific enough to make a decision [134]. © 2022, Jonathan T. Megerian et al.

it was registered on ClinicalTrials.gov (NCT04151290). This study examined device outputs against diagnostic agreements made by two or more independent specialists in a group of children aged 18 to 72 months who had potential developmental delays (425 research participants, 29% with ASD prevalence, 36% female). The device's output PPV was 80.8% (95% confidence intervals (CI), 70.3%–88.8%) for all trial participants, and its NPV was 98.3% (90.6%–100%). The device's sensitivity was 98.4% (91.6%–100%) and specificity was 78.9% (67.6%–87.7%) for the 31.8% of participants who obtained a determinate output (ASD positive or negative). When inputs are too vague to confidently produce a determinate recommendation, the device's indeterminate output serves as a risk control mechanism. The sensitivity

for all study participants would drop to 51.6% (63/122) (95% CI 42.4%, 60.8%) and the specificity to 18.5% (56/303) (95% CI 14.3%, 23.3%) if this risk control measure were eliminated. Experts found that 91% of participants for whom the gadget failed to provide a result had one or more complex neurodevelopmental problems. Regarding the sex, race/ethnicity, income, and educational attainment of the subjects, there were no discernible variations in the device performance. The technology allowed for prompt and very accurate diagnostic evaluation for about one-third of the primary care sample. The tool has the potential to greatly increase the percentage of kids who can receive an ASD diagnosis in a primary care context, which could lead to earlier intervention and more effective utilization of specialized resources.

AI in Mental Healthcare

The discipline of mental health has been slower to adopt AI since this field is more hands-on and patient-centered in their clinical practice than most non-psychiatric practitioners [52]. This field seeks to rely more on “softer” skills, including forming relationships with patients and directly observing patient behaviours and emotions. Mental health clinical data are often in the form of subjective and qualitative patient statements and written notes. However, mental health practice still has much potential benefit to gain from AI technology. AI has great potential to re-define the diagnosis and understanding of mental illnesses. An individual’s unique bio-psycho-social profile is best suited to fully explain his/her holistic mental health. However, there is a relatively narrow understanding of the interactions across these biological, psychological, and social systems. By utilizing AI approaches, it is possible to create risk models and improve pre-diagnosis screening tools that identify a person’s susceptibility to mental illness. Doctors’ time is becoming more and more scarce, and they have to deal with more paperwork and antiquated technology. For mental health professionals, who have to rely on their special human abilities to build therapeutic relationships with their patients and create individualized treatment plans, these issues are especially difficult. Beyond just helping with mental disease diagnosis and detection, AI technology has numerous other advantages [135]. AI algorithms can be used to fully interpret vast and diverse data sets, improve understanding of the prevalence of mental illnesses at the population level, identify biological mechanisms or risk/protective factors, provide technology to track treatment outcomes and/or medication adherence, conduct remote therapy sessions or offer intelligent self-assessments to gauge the severity of mental illness,

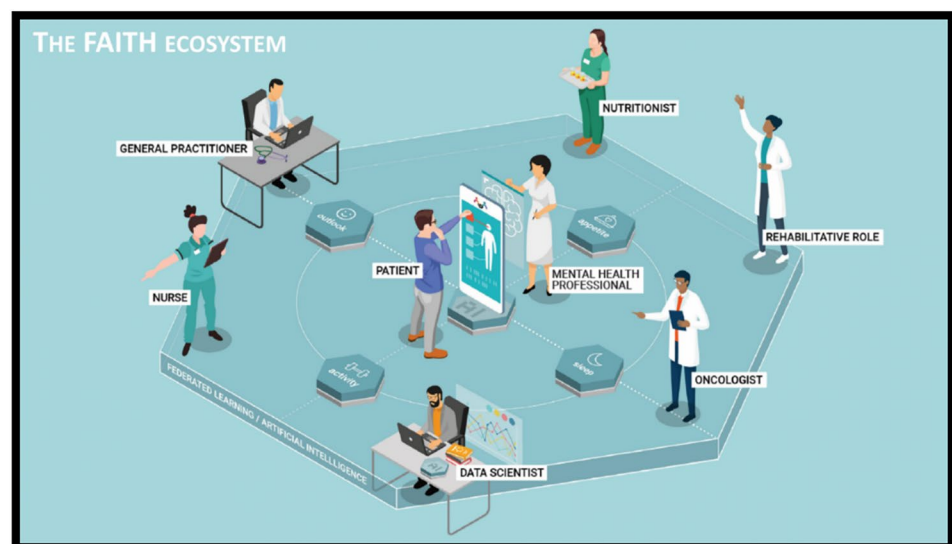
and—perhaps most importantly—allow mental health professionals to concentrate on the human aspects of medicine that can only be accomplished through the clinician-patient relationship [136].

To deploy AI responsibly in mental health, it is critical that algorithms used to predict or diagnose mental health illnesses be accurate and not lead to increased risk to patients. Moreover, those involved in making decisions about the selection, testing, implementation, and evaluation of AI technologies must be aware of ethical challenges, including biased data (e.g., subjective and expressive nature of clinical text data, linking of mental illnesses to certain ethnicities). The FAITH solution as termed was a funded project undertaken in 2020 that utilized a holistic, multidisciplinary approach for cancer care. The FAITH ecosystem for the fight against mental health is shown in Fig. 7.

AI in General Medicine

There are two primary areas of AI application in medicine: virtual and physical components. ML is a representation of the virtual component made up of mathematical algorithms that enhance experience-based learning. Various phases of drug discovery and improvement, including clinical trials, have taken to expand and employ ML algorithms and software to identify novel targets [138]. An example of successes with the virtual component of medicine is the unsupervised protein–protein interaction algorithms that led to novel therapeutic target discoveries. ML in general medicine also enhances awareness of disease and non-disease phenotypes [139, 140], extract high-content material from images at all levels of resolution, develop new biomarkers for prognosis, progression and drug efficacy [141]. Overall, the virtual component has boosted and is still boosting discoveries

Fig. 7 The Ecosystem for the fight against mental health [137]



in genetics and molecular medicine. Physical components like medical devices, including complex robots that assist in providing care (carebots) [142] are also utilized in medicine. The most promising approach is using robots as helpers or companions for the aging population with cognitive deterioration or movement limitations. Robots are used in surgery as assistant surgeons or solo performers [143].

Recently, nanorobots have been developed to address delivery problems that occur when the therapeutic drug diffuses slowly into a target region [10]. A significant obstacle in the use of robots in medical care, aside from ethical concerns, is the clear requirement for a standardized, comparable assessment of the impact of robotic systems on health indicators, as well as measurements of alterations in psychological and physical condition, side effects, and results. As shown in Fig. 8, all stakeholders are needed to share ideas to expand healthcare.

AI/DL for Metastatic Breast Cancer Diagnosis

The microscope has traditionally been the main diagnostic instrument used by pathologists. Manual pathological review is tedious and time-consuming, especially when there are only minor foci of metastatic cancer present in the lymph

nodes, or the lymph nodes are negative for malignancy. Normally, lymph nodes do not have proteins expressed on breast cancer cells, which would increase the sensitivity of cancer metastasis detection. Furthermore, it can be time-consuming and incorrect to identify tiny cancer metastases, even on immunohistochemistry-stained slides. Systems for computer-assisted image analysis have been created to help identify tiny metastatic foci from sentinel lymph node immunohistochemistry slides stained with pancytokeratin [145]. However, these systems are not used clinically. Thus, the development of effective and cost-efficient methods for sentinel lymph node evaluation remains an active area of research, as there would be value to a high-performing system that could increase accuracy and reduce cognitive load at low cost. In several tumor groupings, early cancer diagnosis improves the likelihood of successful treatment. Figure 9 depicts clinical uses of AI in early cancer diagnosis. Workflow triage optimizes diagnostic test quality control, alert systems, and specialist reviews. Image analysis utilizes convolutional neural networks-CNNs (e.g., VNet, UNet) and radiomics for lesion detection and prognosis. Biomarker analysis applies machine learning (ML)/deep learning (DL) techniques on high-dimensional liquid biopsy and whole-genome sequencing data. Recurrence detection/prediction

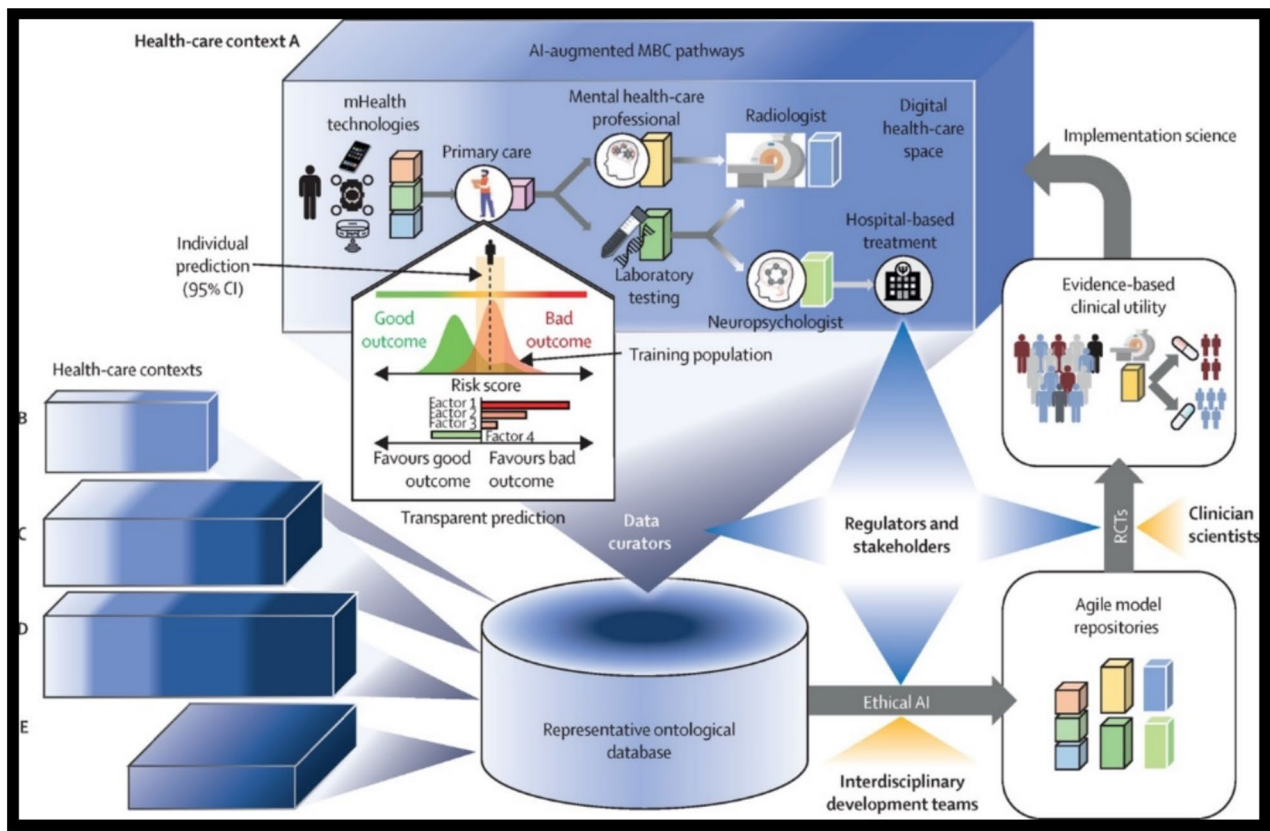


Fig. 8 An AI-informed learning health-care system [144]. © 2022 Koutsouleris et al. Published by Elsevier Ltd.

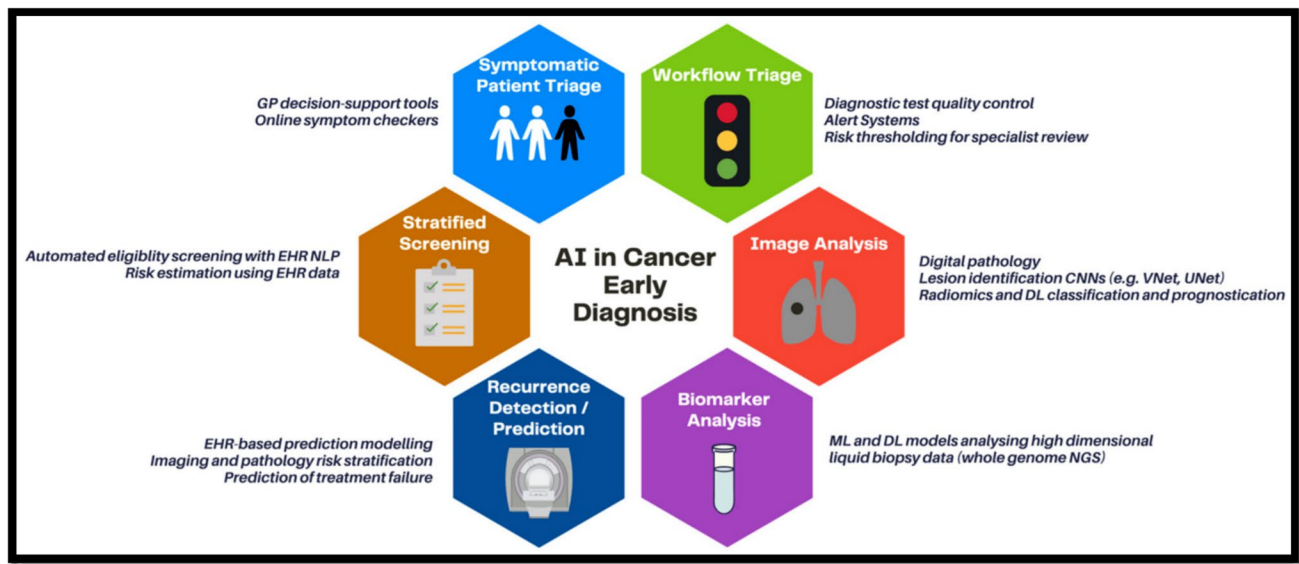


Fig. 9 Clinical uses of AI for early cancer detection. Abbreviations: General practitioners (GPs) and natural language processors (NLPs) EHR stands for electronic health record. Machine learning (ML), deep learning (DL), and next-generation sequencing (NGS) [147]. © CC BY 4.0

combines EHR-based predictive modeling with imaging for treatment outcome forecasting, such as treatment failure prediction and risk stratification. Stratified screening employs automated eligibility screening with an electronic health record (EHR), natural language processors (NLPs), and EHR data for risk estimation of the patient. Symptomatic patient triage leverages General practitioners' (GPs) decision-support tools and online symptom checkers. Using millions of training patches to train a deep neural network (CNN) to create patch-level predictions to distinguish tumor-patches from normal-patches, a DL-based method can be utilized for identifying cancer metastases from entire slide images of breast sentinel lymph nodes [146].

Predictions for the slide-based classification challenge and the tumor-localization task are generated by aggregating the patch-level predictions to generate tumor probability heatmaps, then post-processing these heatmaps. Tissue inside the whole slide imaging (WSI) was recognized and background white space was eliminated to shorten computation times and concentrate investigation on areas of the slide most likely to have cancer metastases. To accomplish this, the team used an automatic threshold-based segmentation technique to identify the background area. The final mask pictures are produced by combining the masks from the H and S channels. Initially, the original image was translated from the RGB colour system to the HSV colour space. Next, the Otsu method is used to calculate the ideal threshold values in each channel [148].

The tissue portions are emphasized by green curves in the detection results. The detection findings indicate that each WSI has an average background region proportion of about

82%. A patch-based classification step and a heatmap-based post-processing stage make up the cancer metastasis detection architecture. In the process of training the model, the patch-based classification step receives as input complete slide images along with the ground truth image annotation, which shows the locations of each WSI's metastatic cancer-containing sections. Figure 10 illustrates how AI, ML, and DL are used in oncology and digital health care to forecast the best course of treatment and resolve medical problems. Digital healthcare is revolutionized by artificial intelligence and its subfields, such as deep learning and machine learning. Digital healthcare is empowering innovations in precision oncology, imaging and digital pathology, next-generation sequencing, drug discovery, and patient data management. The decision making can be improved by AI, whereas the ML and DL enables systems to understand the complex datasets for generating precise predictions. With the integration of AI into digital health systems, it can be served as a powerful tool for the medical professionals in providing a faster diagnosis, and personalized patient specific treatments.

FDA Guidance on AI-Medical Device Applications

The FDA regulates medical devices and approves their market entry majorly through one of three distinct regulatory pathways, De Novo (DEN) review [8] or Premarket Approval (PMA) [9] or Premarket Notification 510(k) clearance. As of right now, there are no particular legal requirements for

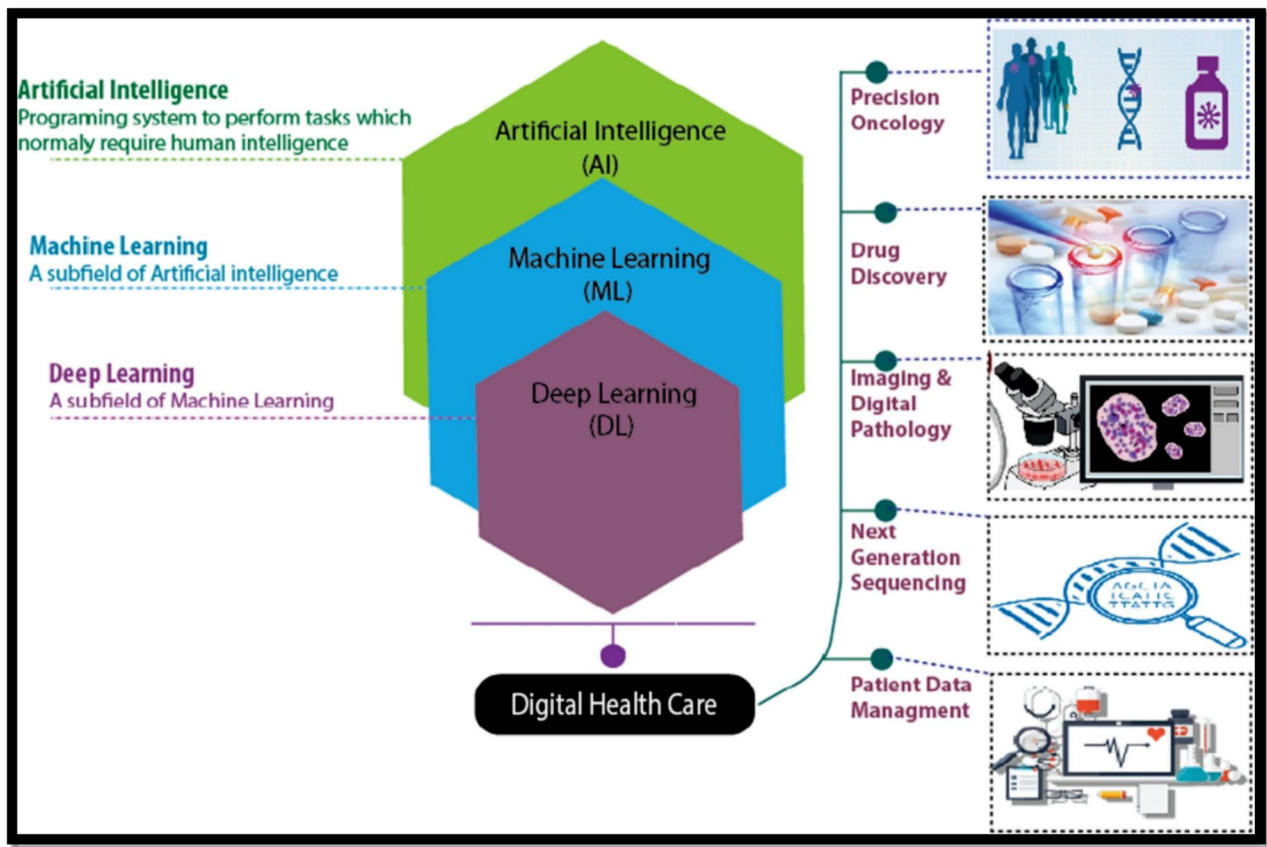


Fig. 10 Applications of AI, ML, and DL in digital health care and oncology to solve healthcare issues and predict optimal treatment outcome [149]. © 2021, Muhammad Javed Iqbal et al.

using AI in medical software or device applications. Specific regulations for AI-enhanced medical software and devices are, nevertheless, rapidly being pushed toward. Nonetheless, given the current regulatory landscape, the FDA still evaluates medical devices and software applications using the conventional criteria (i.e., 510(k), PMA, or De Novo paths). The FDA has published a discussion paper outlining a suggested regulatory framework for premarket submission-required changes to AI/ML-based Software as a Medical Device (SaMD). Viz LVO was the first computer-aided triage software with artificial intelligence to receive FDA clearance through the De Novo premarket review procedure in February 2018. The clinical decision support tool Viz LVO, created by Viz.ai in California, is capable of automatically analyzing CT angiograms for signs of a suspected large artery embolic stroke and sending a mobile device alert to a neurovascular specialist when positive results are found. These stroke patients benefit from prompt treatment, which enhances overall functional outcomes; this has been accomplished by identifying and treating them more quickly thanks to the Viz LVO software. The Viz platform can now be used as a predicate device for future AI-enhanced technologies, thanks to its first-in-class regulatory approval under the FDA

De Novo procedure. Aidoc (Aidoc, Tel Aviv, Israel) was consequently given the go-ahead for clinical use six months later. High acuity CT scans are given priority for radiologist evaluation by this system, which use deep learning algorithms to identify brain hemorrhages. The FDA's ability to scale this paradigm is demonstrated by the quick clearance procedure, which could encourage more innovation in the field. A number of AI-powered clinical decision support software solutions have received approval for U.S. market launch in the previous year. IDx-DR (IDx, Iowa, US) uses artificial intelligence (AI) to automatically analyze retinal images taken with a standard retinal camera and find signs of diabetic retinopathy. Because of its potential to address this unmet need, the device was given breakthrough device status and an expedited access pathway (EAP) prior to approval. The FDA must modify its regulatory paths and clearance procedures to stay up with technological changes, just as it has done in the past with developing medical technology. For instance, the introduction of data-driven, AI-enhanced clinical risk score prediction algorithms has brought out a complicated new regulatory environment. These algorithms could be applied to diagnose or assess the risk of polygenic diseases, provide a more objective and detailed genetic

background for patients, and even direct the treatment decisions and management procedures by informing treatment outcomes based on patient data and predicted therapeutic efficacy. The FDA's approach to regulation will probably depend on the precise responsibilities these unique algorithms will play in the patient's treatment, given the wide range of potential applications of this technology in healthcare. Regulatory requirements for algorithms designed to perform automated decision-making procedures in a healthcare setting are likely to be more stringent than those for algorithms that essentially provide as an extra source of guidance for medical practitioners. It is evident that the use of AI to improve medical software applications has been on the rise. The De Novo pathway has led the software platforms that have been certified for clinical use in these early phases since they lack a substantially equivalent predicate device. A software product can be evaluated and tracked from premarket development to post market performance using the Total Product Lifecycle Regulatory Approach (TPLC) approach, as illustrated in Fig. 11. The approach also allows for ongoing organization-wide excellence demonstration. Figure 12 depicts the schematic representation of the features that should be gathered for FDA-approved AI/ML-enabled medical. The key attributes related to the FDA-approved artificial intelligence-machine

learning enabled medical devices include basic identifiers (device name, type, medical specialty, company name and location), regulatory aspects (regulatory panel details, medical device specialty, approval date, pathway, wait time and decision type), and clinical details (recall history, and performance insights). This all-inclusive framework enables the assessment, tracking, and transparency of AI driven medical devices in healthcare.

AI/ML Approved Medical Devices

FDA's first approved AI-medical device for diagnosis of diabetic retinopathy: diabetic retinopathy condition occurs when high levels of blood sugar leads to damage in the blood vessels of the retina, the light-sensitive tissue in the back of the eye. Diabetic retinopathy is the most common cause of vision loss among the more than 30 million Americans living with diabetes and the leading cause of vision impairment and blindness among working-age adults. Among individuals with diabetes, the prevalence of diabetic retinopathy is approximately 28.5% in the United States [151] and 18% in India [152]. The FDA permitted marketing of the first medical device to use AI to detect diabetic retinopathy (including macular edema) in adults who have diabetes [153].

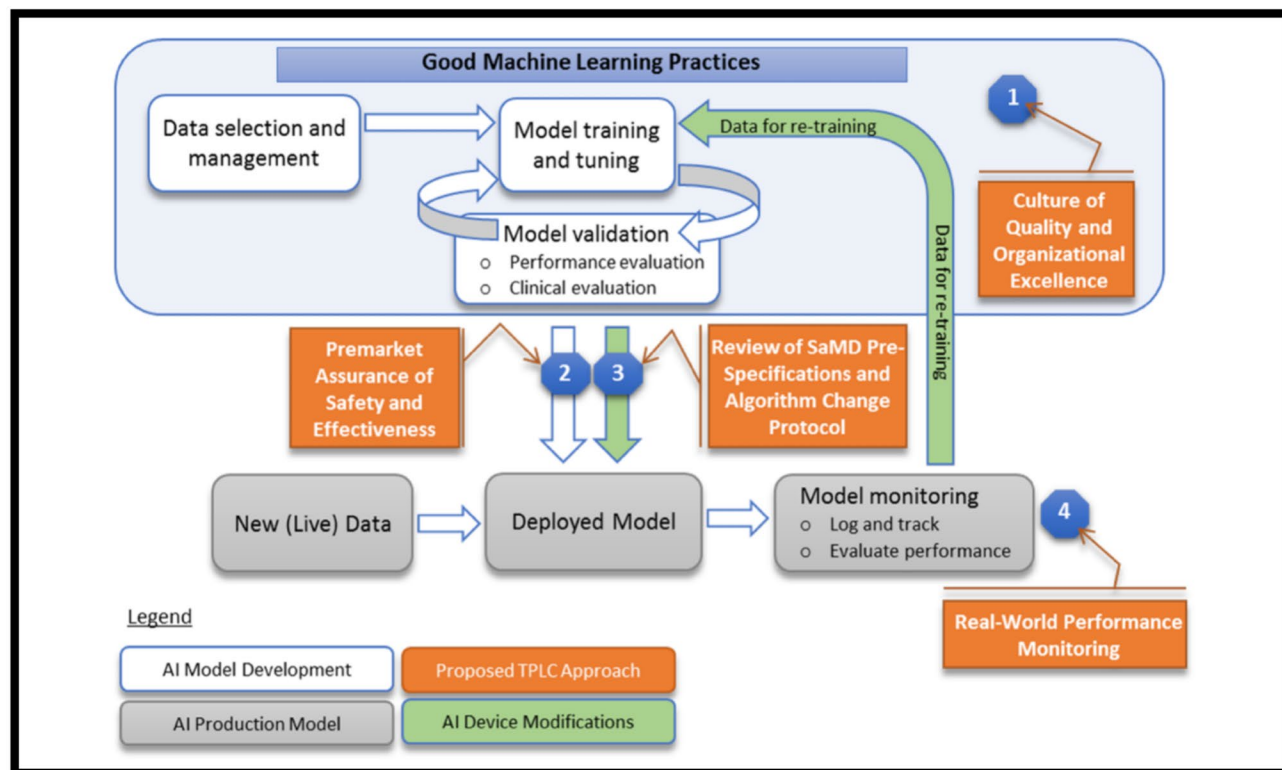


Fig. 11 Overlay of FDA's TPLC approach on AI/ML workflow [154]

Fig. 12 Schematic representation of the medical device features collected for FDA-approved AI/ML-enabled medical devices [150].
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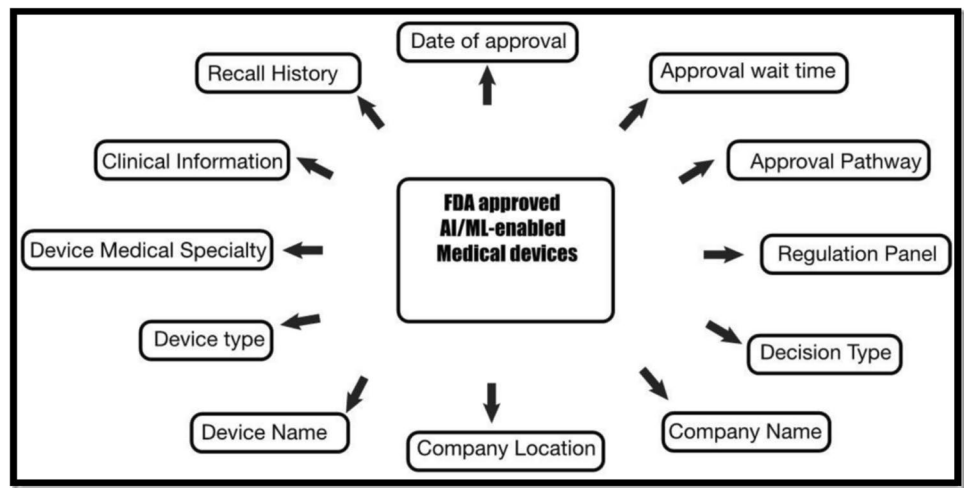


Table 3 Some approved AI/ML-based medical devices in 2024

AI/ML device	Company	Medical specialty
AVIEW CAC	Coreline Soft Co., Ltd	Radiology
NAEOTOM Alpha	Siemens Medical Solutions USA, Inc	Radiology
EyeArt v2.2.0	Eyenuk, Inc	Ophthalmic
AEYE-DS	AEYE Health, Inc	Ophthalmic
CEREC Ortho Software	Dentsply Sirona	Dental

The AI device called IDx-DR works when a trained operator captures two images per eye of a patient using a fundus camera. These images are sent into the IDx-DR and analyze images for signs of diabetic retinopathy (including macular edema), providing results in less than 30 s. When No disease is detected in the result, patience is asked to retest in 12 months, however, if disease is detected, patient is referred to an eye care professional. Some approved AI medical devices in 2024 are shown in Table 3. The AI device called IDx-DR works when a trained operator captures two images per eye of a patient using a fundus camera. These images are sent into the IDx-DR and analyze images for signs of diabetic retinopathy (including macular edema), providing results in less than 30 s. When No disease is detected in the result, patience is asked to retest in 12 months, however, if disease is detected, patient is referred to an eye care professional.

AI-Based Medical Devices in Healthcare: Key Challenges Faced by Innovators

Regulatory and compliance issues: Due to the adaptive nature of AI/ML technologies, the FDA is striving to modernize its regulatory approach for medical devices that are powered by AI. One of the biggest challenges is making

sure that different regulations and standards are followed. **Data security and privacy:** AI-powered medical devices rely on a lot of patient data, which must be safely gathered and handled in order to preserve patient privacy. Data may be used for illegal purposes or may be accessed by unauthorized people. **Maintaining openness and fostering confidence:** It's critical to give precise details about the development, operation, and extensive validation of AI algorithms. To foster trust in these AI-powered gadgets, it is imperative to establish transparent performance measures. AI algorithms need to be tested, updated, and monitored frequently to make sure they continue to function correctly and consistently throughout time and to reduce the possibility of bias and unforeseen effects. **Effectively using AI in clinical practice:** To effectively incorporate AI/ML technology and guarantee patient benefits, clinicians and developers must work closely together. It is imperative that clinicians participate in the software design process to guarantee its efficaciousness and utility. **Taking up moral concerns such as bias:** Among the biggest obstacles to applying AI in healthcare are ethical concerns including data bias and how it affects patient outcomes. It is essential to guarantee patient privacy protection and avoid unexpected repercussions. **Overcoming the reluctance to change:** Healthcare workers and leaders may be reluctant to adopt new AI technologies because they frequently don't think through how these technologies will affect their current workflows and procedures. Successful implementation depends on addressing this reluctance.

Conclusion

Modern healthcare would not be possible without the indispensable role that medical devices play in improving patient outcomes and raising people's quality of life everywhere. Utilization of AI's power in a growing number of devices

and applications is making it possible to organize, classify, investigate, and display healthcare data. AI's impactful algorithms for diagnosis, charting, treatment planning, robotics, education, and other areas are also helping to advance dentistry and the delivery of healthcare as a whole. AI-enabled medical devices are already widely utilized and being quickly adopted. AI developments promise to create a new future where people receive prompt, accurate, and contextualized care and physicians are empowered to make better judgments as digital technologies improve. Even if there are many obstacles in the way of the development and regulation of these devices, industry has proven incredibly resilient and creative in solving these problems. To guarantee the continuous development of healthcare solutions, the medical device sector must continue to be flexible and focused on patient safety and clinical results as technology advances. An overview of AI applications and their effects on healthcare is presented in this review paper intended to introduce the prospects for innovative and collaborative transdisciplinary work. For instance, the innovation of AI has been used in the detection of acute leukemia, breast and pancreatic cancer. Envisaging patient survival rates and estimating patient overall care plan effectiveness are some of the benefits of AI. It may be promising for medical devices to integrate AI, IoT, nanotechnology, biotechnology, customized medicine, and targeted medicines that are customized to the needs of specific individual patients.

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Data Availability Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Competing Interests The author reports there are no competing interest.

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